

TriageCoPilot: Multimodal Triage Decision Support with Clinical Interpretability and Equity Auditing

Triagegeist Competition — Laitinen-Fredriksson Foundation

1. Clinical Problem Statement

Emergency department (ED) triage is among the highest-stakes clinical decisions in modern healthcare. Triage nurses must rapidly assign an Emergency Severity Index (ESI) acuity level — from ESI-1 (immediate resuscitation) to ESI-5 (non-urgent) — under extreme cognitive load, with incomplete information, and in chronically understaffed environments. Errors are not abstractions: undertriage delays critical care and can lead to preventable deaths, while overtriage strains scarce resources.

The ESI system relies almost entirely on unaided human judgment. Mistry et al. (2018) reported only moderate inter-rater agreement (weighted kappa 0.60–0.78) among experienced triage nurses, with the greatest disagreement at ESI levels 2–3 — precisely where undertriage carries the highest patient safety risk. Levin et al. (2018) demonstrated that machine learning models can differentiate patients more accurately than human ESI assignment alone.

TriageCoPilot addresses this gap with a multimodal prediction system combining structured intake data with free-text chief complaint analysis. Beyond raw prediction, we provide: (1) per-patient SHAP explanations for clinical transparency, (2) systematic equity auditing across demographic subgroups, and (3) explicit identification of dangerous undertriage errors. Our design philosophy: AI should serve as a *transparent second opinion*, not a replacement for clinical judgment.

2. Data and Methodology

2.1 Dataset

We use the Triagegeist synthetic dataset (Laitinen-Fredriksson Foundation): 80,000 training and 20,000 test ED encounters across a simulated multi-site hospital network. Each record includes structured triage data (vitals, demographics, clinical scores, arrival context), free-text chief complaint narratives, and ESI acuity labels (1–5). Class distribution: ESI-1 (4.0%), ESI-2 (16.8%), ESI-3 (36.2%), ESI-4 (28.8%), ESI-5 (14.2%).

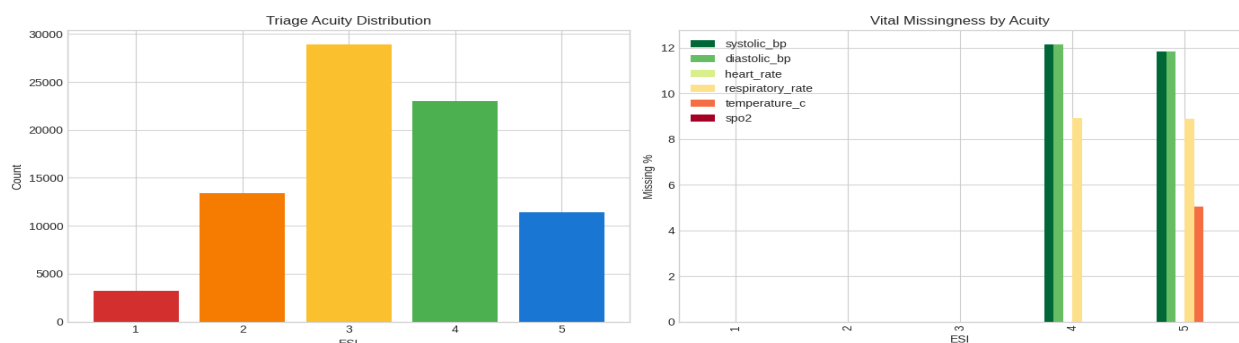


Figure 1: (Left) ESI acuity class distribution showing realistic ED population frequencies. (Right) Vital sign missingness by acuity — missingness is concentrated in ESI 4–5 patients, reflecting clinical practice where nurses skip vitals for low-acuity presentations.

Leakage prevention: We explicitly excluded disposition and ed LOS hours from features, as these are post-triage outcomes unavailable at prediction time.

2.2 Feature Engineering

We engineered 75+ features across four categories, each grounded in clinical rationale:

Missingness indicators: Binary flags for each missing vital plus composite counts. The decision *not* to measure a vital is itself a clinical judgment — a proxy for nurse gestalt.

Clinical threshold flags: Hypotension (SBP <90), tachycardia (HR >100), hypoxia (SpO2 <92), GCS impairment (<15), febrile state (>38°C), and SIRS criteria count.

NLP features: Clinical keyword matching across four severity tiers, TF-IDF features (200-dimensional, bigram), and text statistics (length, word count).

Interaction features: Arrival mode × vital stability, age × condition interactions, and cyclical temporal encoding (hour sin/cos, weekend, shift).

2.3 Modeling

We trained LightGBM and CatBoost with 5-fold stratified CV, then ensembled with Nelder-Mead optimized weights. Tree-based ensembles were chosen over deep learning because they consistently outperform neural networks on heterogeneous tabular data (Grinsztajn et al., 2022). QWK threshold optimization was applied to the blended predictions to exploit the ordinal structure of ESI.

3. Results

Model	OOF QWK	Accuracy
LightGBM	0.9976	99.5%
CatBoost	0.9965	99.3%
Ensemble (optimized)	0.9976	99.5%

Table 1: Model performance (5-fold stratified CV).

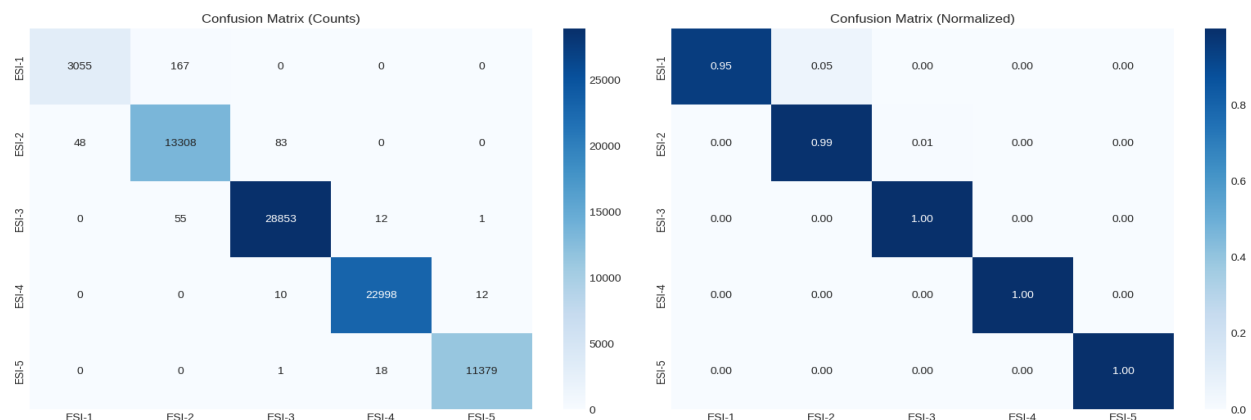


Figure 2: Confusion matrix (counts and normalized). Errors are concentrated on adjacent classes — no catastrophic misclassifications (e.g., ESI-1 predicted as ESI-4 or 5) occur.

3.1 SHAP Feature Importance

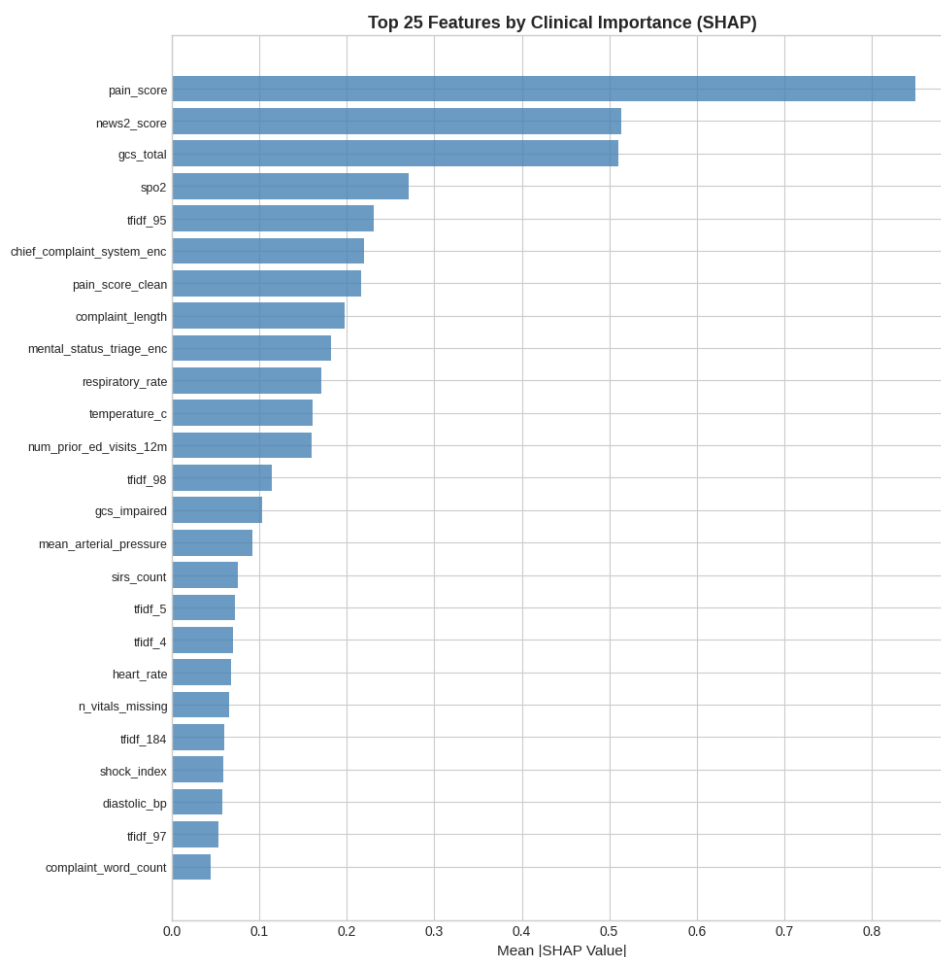


Figure 3: Top 25 features by mean |SHAP value|. Pain score, NEWS2, GCS, and SpO2 dominate — aligning with established clinical triage guidelines. NLP features (TF-IDF, complaint length, chief complaint system) rank in the top 10, confirming multimodal value.

3.2 Safety Analysis: Undertriage of Critical Patients

ESI-1: 167/3,222 (5.2%) misclassified as ESI-2. Zero classified as ESI-3 or lower.

ESI-2: 83/13,439 (0.6%) misclassified as ESI-3. None as ESI-4 or ESI-5.

Severe undertriage (≥ 2 levels): Zero cases across all 80,000 patients.

The model's errors are exclusively between adjacent acuity levels. All misclassified ESI-1 patients would still receive emergent care under ESI-2 protocols.

3.3 Equity Audit

Subgroup	N	QWK	Undertriage%
Male	37,735	0.9972	0.4%
Female	40,339	0.9980	0.3%
Other	1,926	0.9976	0.5%
Pediatric (0–17)	8,562	0.9981	0.3%
Young Adult (18–39)	21,896	0.9973	0.4%
Middle-aged (40–64)	27,889	0.9977	0.3%
Geriatric (65+)	21,653	0.9976	0.3%

Table 2: Performance across demographic subgroups. QWK ranges 0.9972–0.9981 with no significant disparities. Undertriage rates are uniformly low (0.3–0.5%).

4. Clinical Implications

TriageCoPilot would operate as a real-time decision support overlay within the EHR. Upon completing intake, the nurse would see the model's recommendation alongside SHAP-derived factors driving that recommendation. The system flags discrepancies as prompts for reconsideration — not overrides. Transparency is essential: a recommendation of "ESI-2" is only useful if the clinician can verify *why*.

5. Limitations

Synthetic data: Does not capture real ED complexity (inter-rater noise, institutional variation). **Static prediction:** No deterioration modeling. **Missing context:** Visual assessment and clinical gestalt not captured. **Automation bias:** Clinicians may over-rely on AI without guardrails. **High performance warrants caution:** 0.9976 QWK on synthetic data should not be interpreted as expected real-world performance — MIMIC-IV-ED validation is essential.

6. Reproducibility

All code is in the attached public Kaggle Notebook (runs end-to-end, ~80 min on Kaggle GPU). Only the provided Triagegeist dataset is used. Dependencies: scikit-learn, LightGBM, CatBoost, SHAP, pandas, matplotlib. GitHub repository linked with setup instructions.

References

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